



FAKULTÄT FÜR
INFORMATIK

Sensitivity Is Not Fragility:

Decomposing Prompt Effects in Instruction-Tuned and Base Models

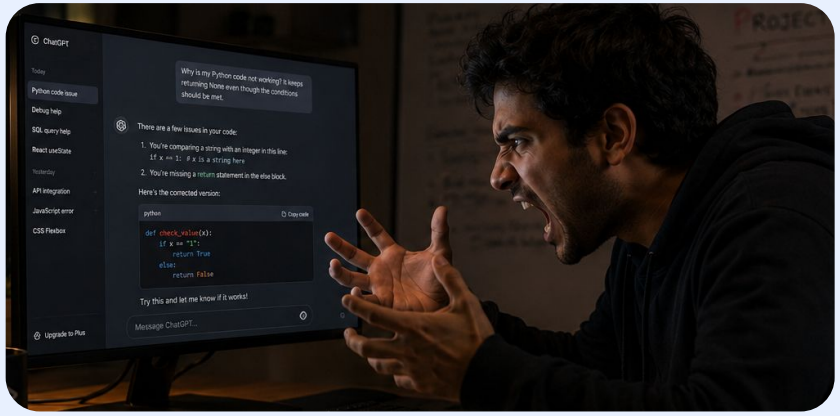
Presentation by:
Mudit Khandelwal, Vishnu Talwar, Neha L Konchady

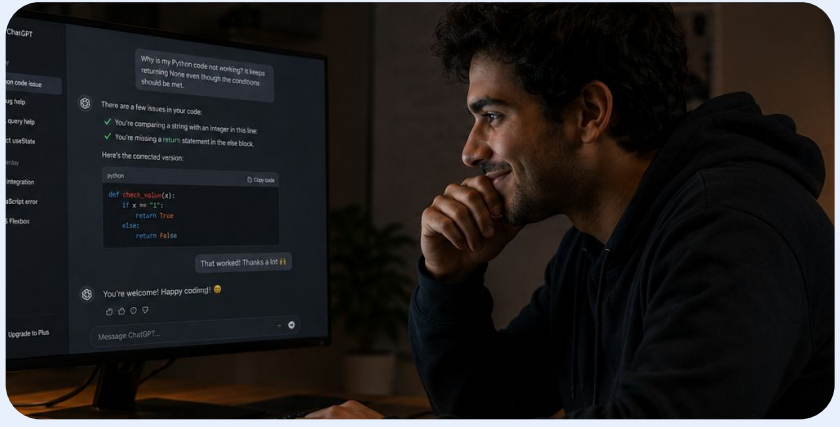
HC-NLP : Group 13

Agenda

- Motivation
- Research Questions
- Models, Datasets & Perturbations
- Evaluation Pipeline & Metrics
- Results
- Base Model Failure Modes
- Sensitivity vs. Fragility
- Statistical Significance
- Conclusion & Limitations


Wait a minute, who are you?


ChatGPT
Why is my Python code not working? It keeps returning None even though the conditions should be met.
There are a few issues in your code:
1. You're comparing a string with an integer in this line:
2. You're missing a return statement in the else block.
Here's the corrected version:
python
Copy code
def check_value(x):
if x == "1":
return True
else:
return False
Try this and let me know if it works!
Message ChatGPT...
Upgrade to Plus

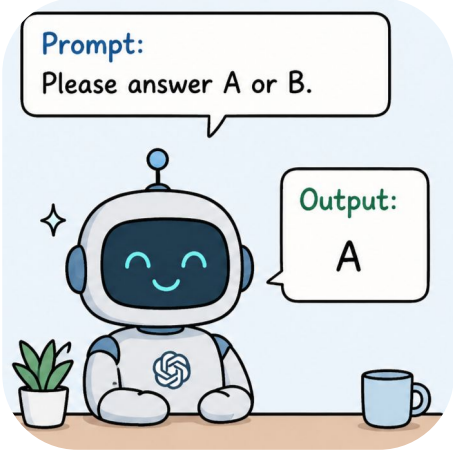

Why is my Python code not working? It keeps returning None even though the conditions should be met.
There are a few issues in your code:
You're comparing a string with an integer in this line:
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Here's the corrected version:
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Copy code
def check_value(x):
if x == "1":
return True
else:
return False
That worked! Thanks a lot 🙌
You're welcome! Happy coding! 😊
Message ChatGPT...
Upgrade to Plus

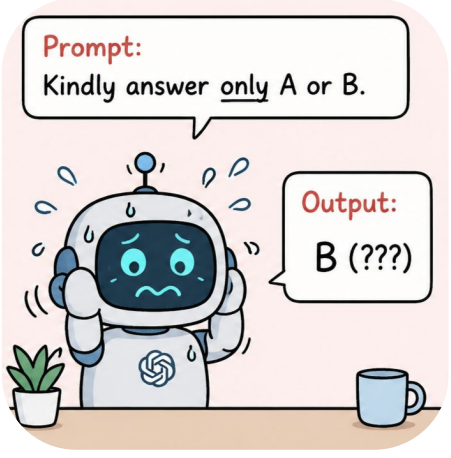
Motivation

Why do small prompt changes matter?

Even when the task remains the same, LLMs can react strongly to small changes in the instruction.

- Small edits in wording, order, or formatting can change accuracy and label behavior.
- Prior work conflates **sensitivity** (how much behavior changes) with **fragility** (harmful brittleness)
- No paired audit across model families + base/instruct variants under a shared protocol





Research Questions



Perturbation effects

RQ1 - How much do minimal instruction perturbations change instruction-tuned model behavior under a shared baseline, and how does this differ across model families and tasks?



Statistical Effects

RQ2 - Which perturbation effects survive paired statistical testing after Holm correction, and which apparent null results are simply underpowered?



Base model failure

RQ3 - Do base models behave as a uniform robustness contrast, or do they exhibit qualitatively distinct failure modes?

Dataset and Models

The Models	The Datasets(850 total examples)
Llama 3.1 (8B) - Instruct & Base variant	Multi-Genre Natural Language Inference (300 items)-MultiNLI
Gemma 2 (9B) – Instruct & Base variant	Big-Bench Hard Date Understanding (250 items) - BBH
Qwen 2.5 (7B) – Instruct & Base variant	AI2 Reasoning Challenge (300 items) - ARC challenge

Premise: The surgeon completed the procedure without any complications and the patient was moved to the recovery ward.

Hypothesis: The operation was successful.

Gold label: A (entailment)

MNLI

Input: Today is Tuesday, March 14, 2023.what day is exactly 3 weeks from today?

Options: (A) April 4, 2023
(B) April 5, 2023
(C) March 28, 2023
(D) April 3, 2023

Gold label: A(April 4, 2023)

BBH

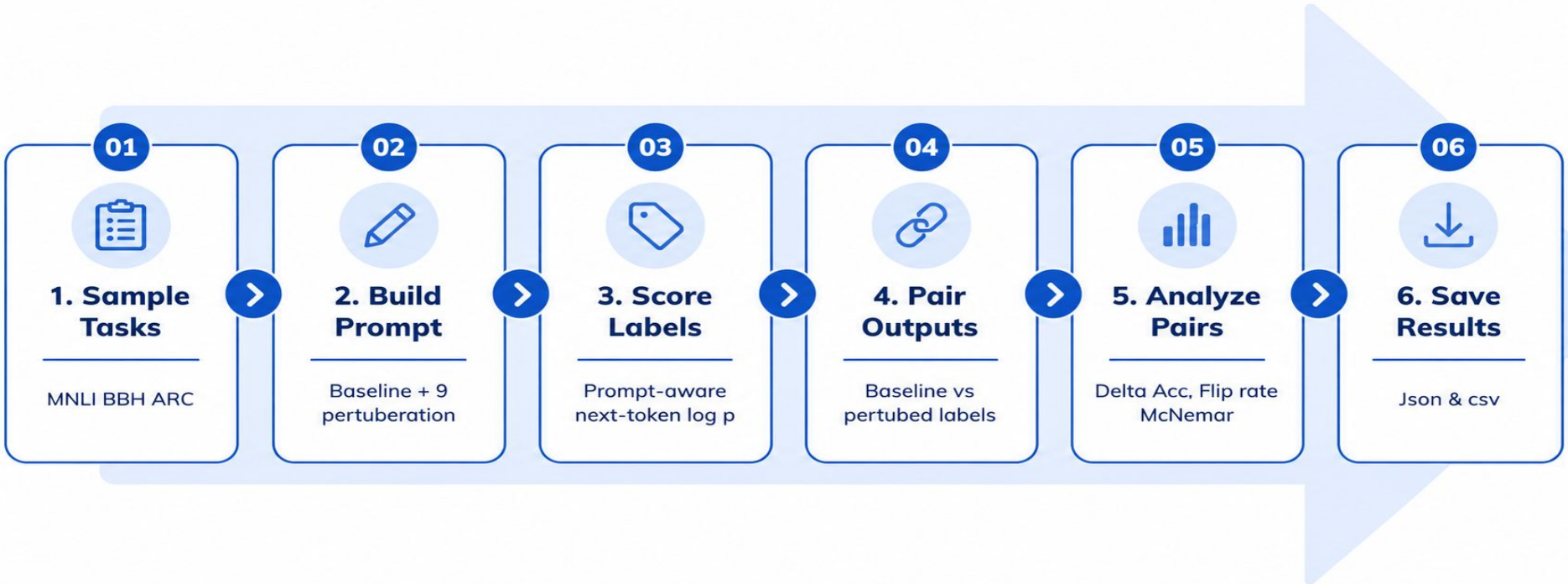
Question: Which of the following is the primary source of energy for almost all life on Earth?

Options: (A) Sunlight
(B) Gravity (C) Wind
(D) Geothermal heat

Gold label: A (Sunlight)

ARC

Evaluation pipeline



Perturbations taxonomy

The Baseline: One shared "drop-in" prompt per task.

The Constraint: Nine *minimal* perturbations that strictly preserve the core task, answer options, and label mapping.

The Goal: Test the prompt interface, *not* to find the optimized prompt.

Perturbation	Operational intent
lex_rewrite	Reword the instruction while preserving label mapping.
lex_typo	Inject realistic character-level typos via nlpaug.
lex_lowercase	Lowercase the instruction; restore answer letters.
struct_inst_first	Move the instruction block before the task body.
ctx_role	Add a role priming preamble.
ctx_irrelevant	Add a semantically irrelevant evaluation note.
fmt_caps	Capitalize the instruction text.
fmt_separators	Replace newlines with visible separators.
instruction_rewrite	Stronger paraphrase keeping the output constraint.

Baseline

MNLI

Premise: The surgeon completed the procedure without any complications and the patient was moved to the recovery ward.
Hypothesis: The operation was successful.
Does the premise entail, contradict, or is it neutral to the hypothesis?
Answer with one letter only: A for entailment, B for neutral, C for contradiction.

Lexical Rewrite

MNLI

Premise: The surgeon completed the procedure without any complications and the patient was moved to the recovery ward.
Hypothesis: The operation was successful.
What is the relationship between the premise and the hypothesis?
Answer with one letter only: A for entailment, B for neutral, C for contradiction.

Lexical Typo

MNLI

Premise: The surgeon completed the procedure without any complications and the patient was moved to the recovery ward.
Hypothesis: The operation was successful.
Deos teh premiss ential, contradict, or is it nuteral to teh hypothesiss?
Answar with one lettar only: A for entailmant, B for neitral, C for contradition.

Lexical Lowercase

MNLI

Premise: The surgeon completed the procedure without any complications and the patient was moved to the recovery ward.
Hypothesis: The operation was successful.
does the premise entail, contradict, or is it neutral to the hypothesis?
answer with one letter only: A for entailment, B for neutral, C for contradiction.

Structure Instruction First

MNLI

Does the premise entail, contradict, or is it neutral to the hypothesis?
Answer with one letter only: A for entailment, B for neutral, C for contradiction.
Premise: The surgeon completed the procedure without any complications and the patient was moved to the recovery ward.
Hypothesis: The operation was successful.

Context Role

MNLI

You are an expert linguist and logician. Premise: The surgeon completed the procedure without any complications and the patient was moved to the recovery ward.
Hypothesis: The operation was successful.
Does the premise entail, contradict, or is it neutral to the hypothesis?
Answer with one letter only: A for entailment, B for neutral, C for contradiction.

Context Irrelevant

MNLI

```
Note: This is part of a larger evaluation suite. Premise: The surgeon completed the procedure without any
complications and the patient was moved to the recovery ward.
Hypothesis: The operation was successful.
Does the premise entail, contradict, or is it neutral to the hypothesis?
Answer with one letter only: A for entailment, B for neutral, C for contradiction.
```

Format Caps

MNLI

```
Premise: The surgeon completed the procedure without any complications and the patient was moved to the recovery
ward.
Hypothesis: The operation was successful.
DOES THE PREMISE ENTAIL, CONTRADICT, OR IS IT NEUTRAL TO THE HYPOTHESIS?
ANSWER WITH ONE LETTER ONLY: A FOR ENTAILMENT, B FOR NEUTRAL, C FOR CONTRADICTION.
```

Format Separators

MNLI

```
Premise: The surgeon completed the procedure without any complications and the patient was moved to the recovery
ward.|Hypothesis: The operation was successful.|Does the premise entail, contradict, or is it neutral to the
hypothesis?|Answer with one letter only: A for entailment, B for neutral, C for contradiction.
```

Instruction Rewrite

MNLI

```
Premise: The surgeon completed the procedure without any complications and the patient was moved to the recovery
ward.
Hypothesis: The operation was successful.
Determine the relation between the premise and the hypothesis.
Answer with one letter only: A for entailment, B for neutral, C for contradiction.
```

Instruction Rewrite

BBH

```
Today is Tuesday, March 14, 2023. Jane's birthday is exactly 3 weeks from today. What is the date of Jane's
birthday?
(A) April 4, 2023
(B) April 5, 2023
(C) March 28, 2023
(D) April 3, 2023    Select the correct option. Write only the corresponding letter.
```

Instruction Rewrite

ARC

```
Question: Which of the following is the primary source of energy for almost all life on Earth?
(A) Sunlight
(B) Gravity
(C) Wind
(D) Geothermal heat
Which option is correct? Write just the letter.
```

Metrics and Scoring

Δacc	Acc_{base} - Acc_{pert}: pp; positive = harm, negative = improvement
Flip Rate (FR)	Flip rate: fraction of examples where prediction changes
Sensitivity Score (SS)	FR × |Δacc|: bounded composite; requires *both* many flips and large accuracy shift
Semantic distance (d□e□)	Cosine distance between SBERT embeddings of baseline vs. perturbed instruction

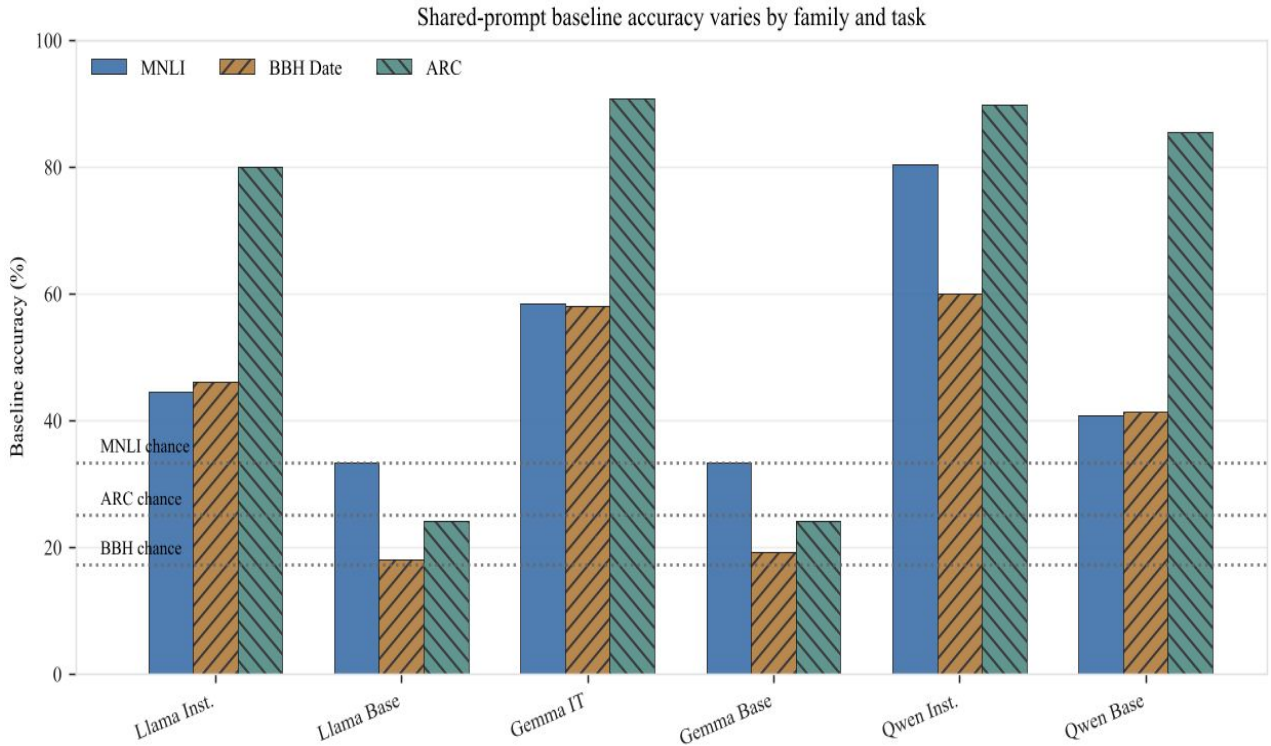
Scoring: Constrained letter outputs scored via next token log probabilities; chat-template continuation validation for instruct models

The Before: Baseline accuracy

Instruct models generally outperform base models under the shared prompt.

Qwen Instruct gives the strongest overall baseline, especially on MNLI.

Qwen Base is unusually strong on ARC, so its later drop is not just “base models are weak.”



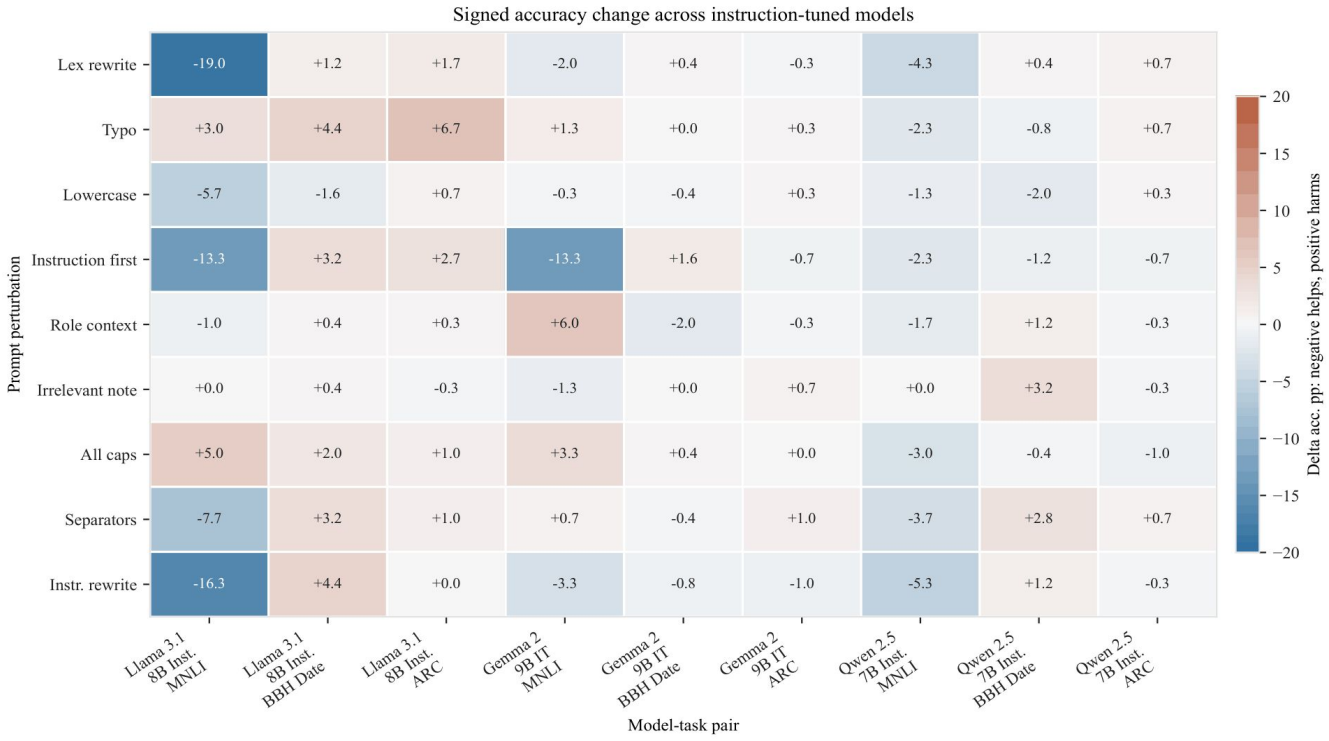
The After: Where Prompt Changes Help vs. Hurt

Large prompt effects exist, but they are model and task specific.

MNLI shows the largest improvements, especially for Llama Instruct and Gemma Instruct.

Positive Δacc values indicate harm; negative values indicate recovery.

Therefore, sensitivity must be interpreted with direction, not only magnitude.



Blue = improvement, red = harm. Positive Δacc means the perturbation hurt accuracy

Llama on lex_rewrite

Task: ARC | Model: llama3.1-8b-instruct	
Base Accuracy	80.00%
Perturbed Accuracy	78.3%
Δacc	+1.7pp
FR	0.040
SS	0.07
d□e□	0.2872

Task: MNLI | Model: llama3.1-8b-instruct	
Base Accuracy	44.33%
Perturbed Accuracy	63.3%
Δacc	-19pp
FR	0.403
SS	7.7
d□e□	0.1114

Task: BBH | Model: llama3.1-8b-instruct	
Base Accuracy	46.0%
Perturbed Accuracy	44.8%
Δacc	+1.2pp
FR	0.028
SS	0.03
d□e□	0.068

Same perturbation, completely different outcomes - sensitivity is a model × task × perturbation interaction.

Δacc: base accuracy - Pert accuracy FR: Flip Rate SS: Sensitivity Score d□e□: Semantic Distance

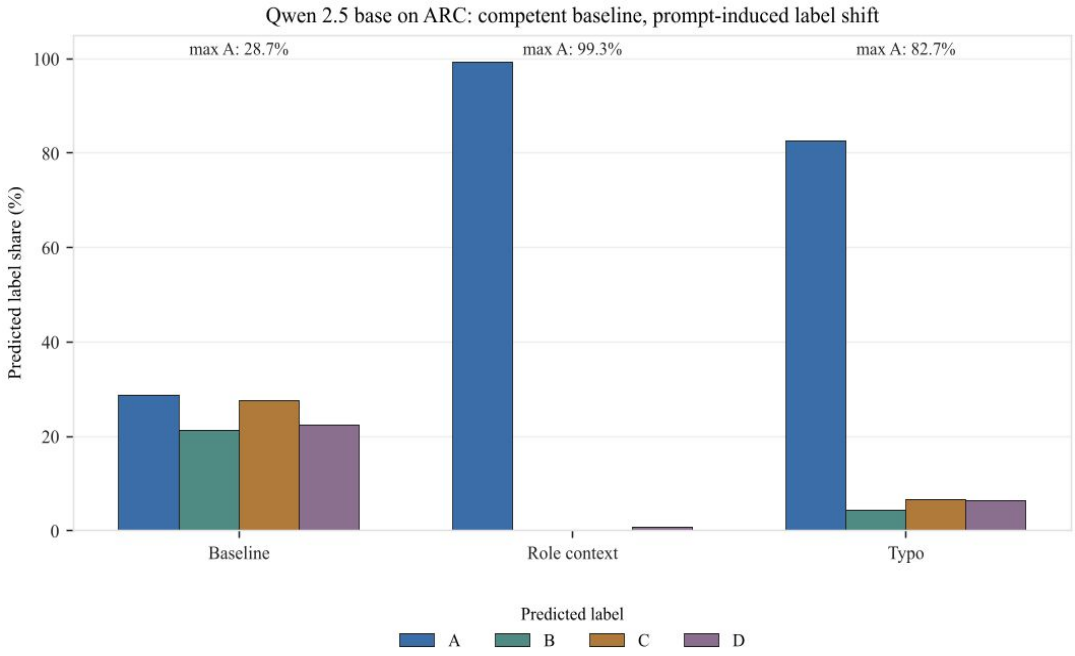
Case Study 1: Llama Inst. on MNLI recovers from a weak baseline

Condition	MNLI accuracy	Meaning
Baseline	44.3%	Weak prompt model fit
lex_rewrite	63.3%	-19.0 pp improvement
instruction_rewrite	60.7%	-16.3 pp improvement
struct_inst_first	57.7%	-13.3 pp improvement

This is sensitivity, but not fragility: the prompt change improves behavior.

Case Study 2: Qwen Base collapses on ARC

Condition	ARC accuracy	Label behavior
Baseline	85.3%	Balanced A/B/C/D predictions
ctx_role	24.7%	Almost all predictions become A
lex_typo	40.7%	Strong skew toward A



This is genuine fragility: high flip rate + harmful Δacc + label collapse.

Case Study 3: Base models are not one phenomenon

Base models do not behave as a clean robustness control group.

Base-model behavior	Example	Interpretation
Non-engagement near chance	Llama Base, Gemma Base	Low sensitivity does not mean robustness
Competent baseline	Qwen Base on ARC, BBH, MNLI	Can solve the task under the shared prompt
Prompt-induced collapse	Qwen Base + ctx_role for ARC	Small prompt change causes degenerate label behavior

Base-model results are useful as case studies, but not as a clean robustness comparison.

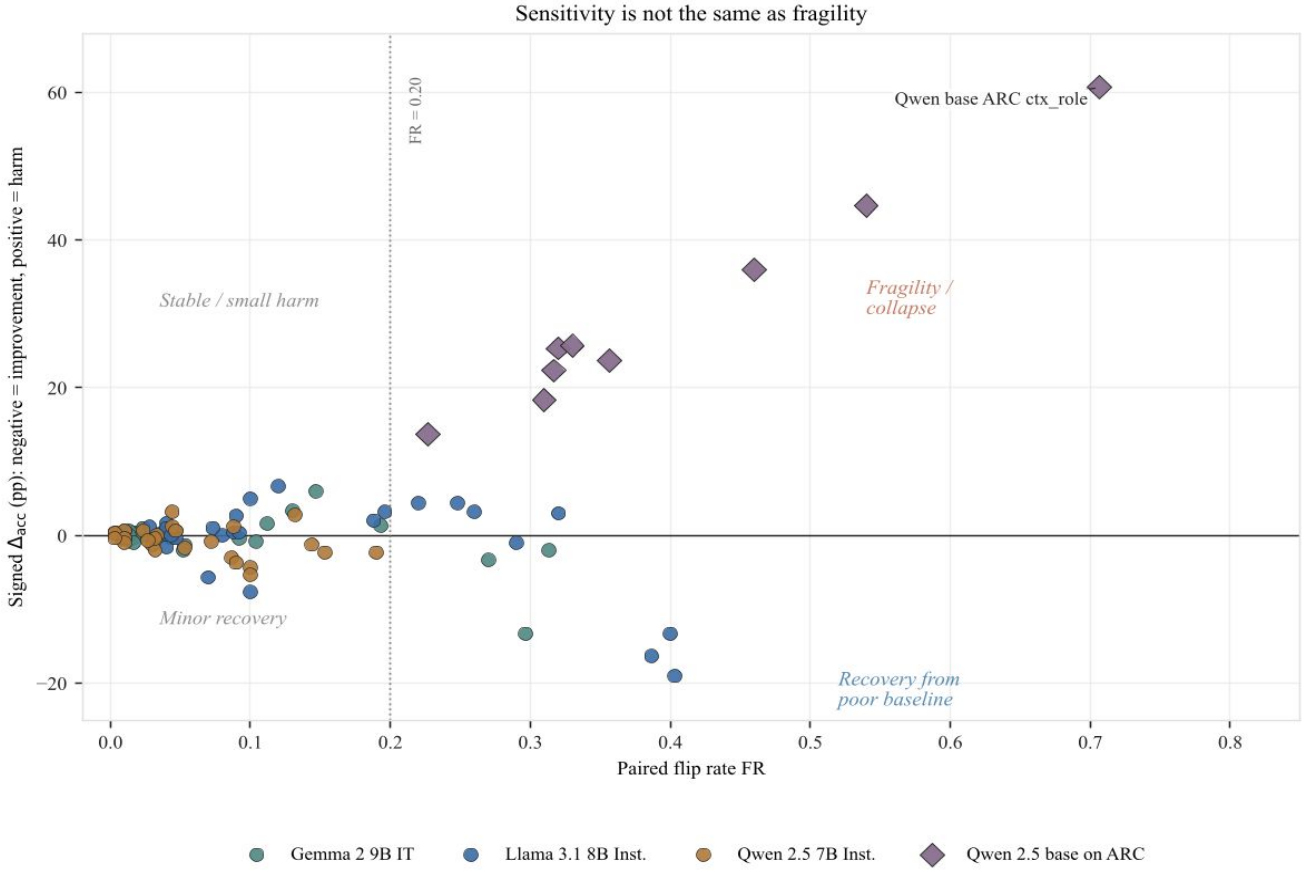
Mapping Recovery and Collapse

Recovery from poor baseline : High FR + large negative Δacc (improvement). Llama-MNLI cluster.

Fragility / collapse: High FR + large positive Δacc (harm). Qwen base ARC ctx_role: accuracy 85.3% to 24.7%, FR = 0.71

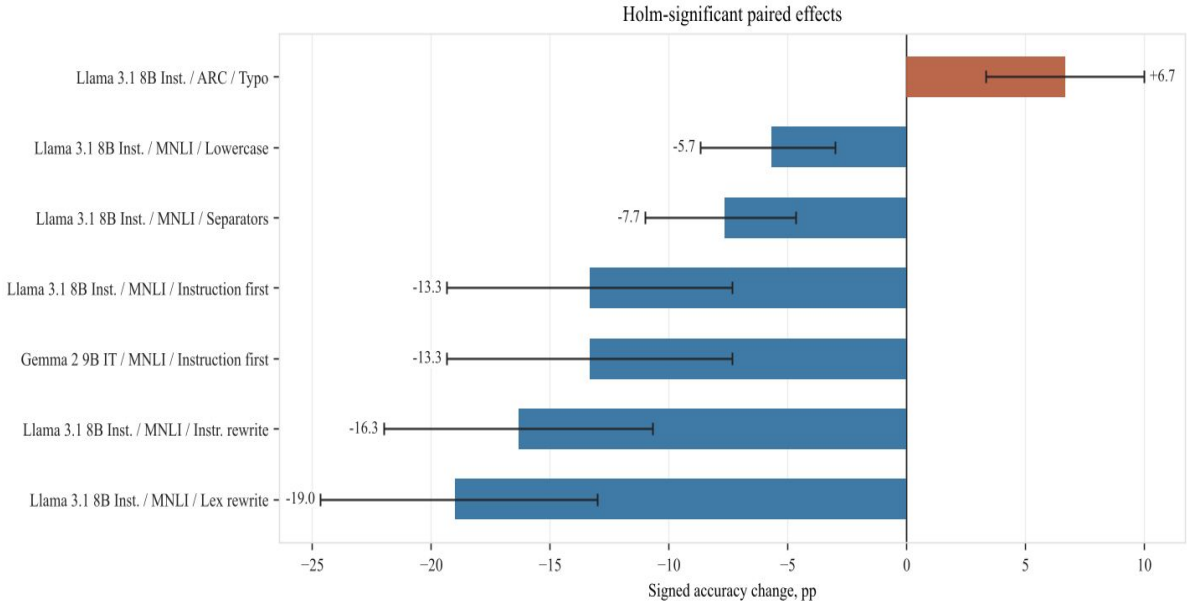
Stable / small harm: Low FR + small positive Δacc. Most BBH and ARC effects.

Minor recovery: Low FR + small negative Δacc. Qwen-Instruct across all tasks.



Statistical Significance

95% CI	McNemar's Test	Holm-Bonferroni correction
Estimates the plausible range of the true accuracy drop under resampling providing a measure of uncertainty	Whether the perturbation changes predictions asymmetrically on the same examples by comparing correct→wrong vs wrong→correct swaps (b vs c)	Adjusts p-values across multiple perturbations to control the family-wise error rate, reducing false positives when many tests are run together.



- Only 7 of 81 instruction model perturbation effects survive Holm correction.
- 6 of 7 are MNLI improvements.
- Only one significant harmful instruction tuned effect: Llama ARC under typos.
- 52 of 81 tests are underpowered due to fewer than 20 discordant pairs, null result here means inconclusive, not zero effect.

Conclusion

1. Prompt sensitivity is real, but it is not automatically fragility.

2. A high flip rate only becomes concerning when it also causes a harmful accuracy drop.

3. Some perturbations reveal collapse; others recover from a poor baseline prompt.

4. Robust evaluation should report multiple prompts, signed Δacc, flip rate, and label diagnostics.

Limitation and Future Work

1. Shared baseline vs. model optimized prompts Cross-model sensitivity differences entangled with baseline prompt model fit
2. Single seed and subset Results are reproducible but do not estimate sampling variance
3. Quantization confound 4-bit NF4 may itself alter calibration and sensitivity; full precision comparison not done
4. Limited perturbation suite (9 items, unbalanced categories) Category level averages would be misleading; category comparisons avoided
5. Asymmetric chat templates Qwen auto-injects system message; Llama/Gemma do not confounds cross model comparison
6. Holm correction is conservative Many BBH/ARC tests underpowered; null = inconclusive, not zero effect

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THANK YOU